

LAB # 05a

Construct if else statements in java

Objective

Implementation of various decisions making constructs.

Lab Goals

To provide working insight into the usage of decision making statements using the following:

- a. If statement.
- b. If then else statement.
- c. Nested If statement.

Introduction to Decision making statements

Decision making statements contain conditions that are evaluated by the program. If the condition is true, then a set of statements are executed and if the condition is false then another set of statements is executed. ... Otherwise, statements in the body of else are executed.

If statement

If statement is the simplest decision making statement. It is used to decide whether a certain statement or block of statements will be executed or not.

Syntax

```
if (condition) {  
    // Statements to execute if  
    // condition is true  
}
```

Example#1

```
public class Main {  
    public static void main(String[] args) {  
        int x = 10;  
        if (x > 5) { System.out.println("x is greater than 5"); }  
    }  
}
```

Output

```
x is greater than 5
```

Example#2

```
public class Main {  
    public static void main(String[] args) {  
        String str = "Hello";  
        if (!str.isEmpty()) {  
            System.out.println("The string is not empty");  
        }  
    }  
}
```

Output

```
The string is not empty
```

Example#3

```
public class Main {  
    public static void main(String[] args) {  
        char ch = 'a';  
        if (ch == 'a' || ch == 'e' || ch == 'i' || ch == 'o' || ch == 'u') {  
            System.out.println("The character is a vowel");  
        }  
    }  
}
```

Output

```
The character is a vowel
```

If-else Statement

If statement alone tells us that if a condition is true it will execute a block of statements and if the condition is false it won't. But what if we want to do something else if the condition is false. Here comes the else statement.

Syntax

```
if (condition) {  
    // Code block to execute if the condition is true  
} else {  
    // Code block to execute if the condition is false  
}
```

Example#1

```
public class Main {  
    public static void main(String[] args) {  
        int y = 3;  
        if (y % 2 == 0) {  
            System.out.println("y is even");  
        } else {  
            System.out.println("y is odd");  
        }  
    }  
}
```

Output

```
y is odd
```

Example#2

```
public class Main {  
    public static void main(String[] args) {  
        int year = 2024;  
        if ((year % 4 == 0 && year % 100 != 0) || (year % 400 == 0)) {  
            System.out.println(year + " is a leap year");  
        } else {  
            System.out.println(year + " is not a leap year");  
        }  
    }  
}
```

Output

```
2024 is a leap year
```

if-else-if ladder

An if-else-if ladder is a programming construct used to execute one of several blocks of code based on multiple conditions. It allows you to test multiple conditions sequentially and execute a different block of code based on which condition evaluates to true.

Syntax

```
if (condition1) {  
    // code to execute if condition1 is true  
} else if (condition2) {  
    // code to execute if condition1 is false and condition2 is true  
} else if (condition3) {  
    // code to execute if condition1 and condition2 are false and condition3 is true  
} else {  
    // code to execute if all conditions are false  
}
```

Example#1

```
public class Main {  
    public static void main(String[] args) {  
        int a = 15, b = 20, c = 10;  
        if (a >= b && a >= c) {  
            System.out.println("a is the largest number");  
        } else if (b >= a && b >= c) {  
            System.out.println("b is the largest number");  
        } else {  
            System.out.println("c is the largest number");  
        }  
    }  
}
```

Output

```
b is the largest number
```

Example#2

```
public class Main {  
    public static void main(String[] args) {  
        int marks = 75;  
        if (marks > 90) {  
            System.out.println("Grade A");  
        } else if (marks > 80) {
```

```
        System.out.println("Grade B");
    } else if (marks > 70) {
        System.out.println("Grade C");
    } else if (marks > 60) {
        System.out.println("Grade D");
    } else {
        System.out.println("Grade F");
    }
}
}
```

Output

Grade C

Example#3

```
public class Main {
    public static void main(String[] args) {
        int x = 2, y = 3;
        if (x > 0 && y > 0) {
            System.out.println("Point lies in the first quadrant");
        } else if (x < 0 && y > 0) {
            System.out.println("Point lies in the second quadrant");
        } else if (x < 0 && y < 0) {
            System.out.println("Point lies in the third quadrant");
        } else if (x > 0 && y < 0) {
            System.out.println("Point lies in the fourth quadrant");
        } else {
            System.out.println("Point lies on one of the axes");
        }
    }
}
```

Output

Point lies in the first quadrant

Nested If

A nested if is an if statement that is the target of another if or else. Nested if statements mean an if statement inside an if statement. Yes, java allows us to nest if statements within if statements.

Syntax

```
if (condition1) {  
    // Executes when condition1 is true  
    if (condition2)  
    {  
        // Executes when condition2 is true  
    }  
}
```

Example#1

```
public class Main {  
    public static void main(String[] args) {  
        int a = 30;  
        if (a > 0) {  
            if (a % 2 == 0) {  
                System.out.println("a is a positive even number");  
            } else {  
                System.out.println("a is a positive odd number");  
            }  
        } else {  
            System.out.println("a is not a positive number");  
        }  
    }  
}
```

Output

```
a is a positive even number
```

Example#2

```
public class Main {  
    public static void main(String[] args) {  
        int age = 25;  
        boolean isMember = true;  
  
        if (age >= 60) {  
            if (isMember) {
```

```
        System.out.println("You are eligible for a senior discount");
    } else {
        System.out.println("You are eligible for a senior discount with membership");
    }
    } else {
        System.out.println("You are not eligible for a senior discount");
    }
}
}
```

Output

```
You are not eligible for a senior discount
```

Example#3 :: Sorting three numbers in ascending order

```
public class Main {
    public static void main(String[] args) {
        int num1 = 5, num2 = 10, num3 = 3;
        if (num1 <= num2) {
            if (num1 <= num3) {
                System.out.print(num1 + ", ");
                if (num2 <= num3) {
                    System.out.print(num2 + ", " + num3);
                } else {
                    System.out.print(num3 + ", " + num2);
                }
            } else {
                System.out.println(num3 + ", " + num1 + ", " + num2);
            }
        } else {
            if (num2 <= num3) {
                System.out.print(num2 + ", ");
                if (num1 <= num3) {
                    System.out.print(num1 + ", " + num3);
                } else {
                    System.out.print(num3 + ", " + num1);
                }
            } else {
                System.out.println(num3 + ", " + num2 + ", " + num1);
            }
        }
    }
}
```

Output

3, 5, 10

Exercise

- A. Take values of length and breadth of a rectangle from user and check if it is square or not.
- B. A school has following rules for grading system:

a.	Below 25	F
b.	25 to 45	E
c.	45 to 50	D
d.	50 to 60	C
e.	60 to 80	B
f.	Above 80	A

Write a program to enter marks and print the corresponding grade.

- C. A student will not be allowed to sit in exam if his/her attendance is less than 75%. Take following input from user, Number of classes held, Number of classes attended and print percentage of class attended. Is student is allowed to sit in exam or not.
- D. **Create a console-based Registration Form using Java.**

Your program should ask the user to enter the following details:

1. Full Name
2. Age
3. Email
4. Password

Perform **validation using an if-else if-else ladder**:

- Full Name: Should not be empty.
- Age: Must be between 18 and 60.
- Email: Must contain @ and a . after @.
- Password: Must be at least 6 characters long.

Display a message indicating whether registration is **successful** or what the validation error is.